EpiHack Participatory Surveillance Platform

LLM-Ready User Story Prompts

One Health Integration | Critical Analysis | Community Engagement

Prepared: May 2026 | EpiHack Data Science Team

Document Overview

This document presents a critical evaluation of 10 prototype prompt templates from the EpiHack contextual risks framework, followed by 40 LLM-ready user story prompts structured for deployment in a One Health participatory surveillance platform. Section 1 provides a systematic critical analysis. Section 2 contains 20 prompts for individual survey participants. Section 3 contains 20 prompts for public health officials. Section 4 addresses additional implementation considerations. Section 5 presents 15 clarifying questions for platform development.

Each user story follows the standard format: User Story (As a [role], I want [capability], so that [outcome]) · LLM Prompt Template (complete, parameterized instruction) · Objective · Data Sources & Priority Level.

Section 1: Critical Analysis of Source Prompt Templates

The EpiHack contextual risks document presents 10 prototype prompt templates organized around ten epidemiological narrative patterns. The following analysis evaluates each template against best practices for LLM-ready participatory surveillance prompts, applying a One Health lens and considering the operational realities of Arizona's public health infrastructure.

1.1 Strengths of the Source Templates

- User-centric narrative framing: The 'As a participant…' story structure establishes relatable contexts, lowering the cognitive barrier for community engagement.
- Temporal specificity: Seven of ten templates explicitly reference time windows (e.g., past N days, 30-day baseline), which is essential for syndromic surveillance signal detection.
- Symptom-level granularity: Templates address individual symptom flags rather than aggregate illness categories, enabling fine-grained epidemiological parsing.
- Geographic sensitivity: Postal code and community-level framing aligns with Arizona's diversity of rural, tribal, and urban geographies.
- Re-engagement design: The re-engagement and reporter density templates recognize the operationally critical problem of survey attrition in longitudinal surveillance cohorts.

1.2 Identified Gaps and Limitations

- Absence of One Health integration: None of the 10 templates references animal, environmental, or vector-borne data streams, despite the platform's stated One Health scope. This is a critical omission for zoonotic and climate-sensitive disease surveillance.
- No equity stratification: The Arizona Social Vulnerability Index (AZSVI) is not referenced in any template. Demographic shift templates should be stratified by vulnerability quintile.
- Insufficient uncertainty communication: Templates do not instruct the LLM to communicate statistical uncertainty (e.g., small cell sizes, low response rates), which risks overconfident outputs.
- Missing clinical triage guidance: No template directs users toward clinical decision support or emergency referral pathways when symptom burden is severe.
- Lack of tribal data sovereignty framing: Arizona has 22 federally recognized tribes. Templates do not acknowledge data governance boundaries or culturally appropriate messaging.

- No feedback loop closure: Templates generate outputs but do not prompt the LLM to explain how participant-reported data influenced public health action, undermining participant trust.
- LLM prompt templates are underspecified: Message templates lack system-role instructions, output format constraints, and chain-of-thought directives needed for reliable LLM performance.
- No multilingual or health-literacy scaffolding: Arizona's population includes significant Spanish-speaking and low-health-literacy communities; templates assume English proficiency and standard reading comprehension.

1.3 Expanded Objective Framework

An augmented prompt taxonomy for EpiHack should address six functional objectives: (1) Personal Health Intelligence — informing participants about their own risk trajectory; (2) Community Epidemiological Awareness — situating individual reports within community-level patterns; (3) One Health Signal Integration — connecting human symptom reports with animal, environmental, and vector data; (4) Equity-Stratified Risk Communication — ensuring vulnerable populations receive appropriately tailored messaging; (5) Clinical Decision Support — bridging surveillance data with actionable clinical guidance; and (6) Public Health Operational Intelligence — equipping officials with actionable signals for outbreak response, resource allocation, and policy decisions.

Section 2: LLM-Ready User Story Prompts — Individual Survey Participants

The following 20 user story prompts are designed for participatory surveillance participants. Each prompt is structured as a complete LLM instruction suitable for integration into the EpiHack platform backend. Prompts assume access to the participant's longitudinal survey record and relevant community-level aggregates.

1. Personal Symptom Trajectory	
User Story	As a survey participant, I want to see how my reported symptoms have changed over the past 30 days compared to my earlier baseline, so that I can understand whether my health is improving or deteriorating.
LLM Prompt	You are a public health data interpreter. The participant has reported the following symptoms over the past 30 days: {symptom_timeseries}. Their 30-day prior baseline was: {baseline_summary}. In 3–4 sentences, describe the participant's symptom trajectory in plain language, note any symptoms that have newly appeared or resolved, and recommend whether they should consult a healthcare provider. Do not diagnose. Use a supportive, non-alarming tone.
Objective	Provide personalized longitudinal symptom feedback to improve health self-awareness and promote timely care-seeking behavior.
Data Sources / Priority	Sources: EpiHack survey record, fakeData.csv symptom flags, Date of Illness field | Priority: HIGH

2. Community Comparison	
User Story	As a survey participant, I want to know how my symptoms compare to others in my ZIP code area, so that I can understand whether I may be experiencing something that is circulating locally.
LLM Prompt	You are a community health communicator. The participant reports {participant_symptoms} and lives in postal area {postal_code}. In their community, {community_symptom_summary} (% of reporters with each symptom in the past 7 days). In 2–3 sentences, explain whether the participant's symptoms are consistent with community trends, and advise them on any precautions. Avoid stigmatizing language. Do not speculate about specific diseases.
Objective	Contextualize individual symptom reports within community-level syndromic patterns to promote informed health decision-making.
Data Sources / Priority	Sources: EpiHack aggregate rates by postal code, baseline_change_report outputs | Priority: HIGH

3. Emerging Symptom Alert	
User Story	As a survey participant, I want to be notified when a symptom I am experiencing is newly emerging in my community, so that I can take timely precautions.
LLM Prompt	You are a public health alert system. The participant reported {new_symptom} today. Community surveillance data shows this symptom has shifted from 0% to {current_rate}% prevalence in postal area {postal_code} in the past 7 days (signal: NEW). Draft a 2–3 sentence alert message for the participant explaining the emerging pattern, recommending practical precautions (hygiene, ventilation, masking), and advising them to monitor symptoms. Do not alarm unnecessarily.
Objective	Generate timely, actionable early-warning notifications for participants when their reported symptoms coincide with emerging community signals.
Data Sources / Priority	Sources: baseline_change_report.py signal='new', EpiHack alert banner logic | Priority: CRITICAL

4. Heat-Related Risk Contextualization

User Story	As a survey participant living in Arizona, I want to understand whether my symptoms could be related to extreme heat exposure, so that I can take appropriate protective action.
LLM Prompt	You are a One Health environmental health advisor. The participant reported {symptoms} on {report_date}. The NWS HeatRisk index for {postal_code} on that date was {heat_risk_level} (scale: 0–4). In 3–4 sentences, explain the relationship between heat exposure and the reported symptoms, describe signs of heat illness requiring emergency care, and recommend protective behaviors. Cite the heat risk level explicitly.
Objective	Integrate environmental heat data with participant symptom reports to deliver climate-sensitive health guidance within the One Health framework.
Data Sources / Priority	Sources: NWS HeatRisk API, HEAT.AZ.gov, NIHHIS, EpiHack symptom flags (Fever, Muscle Aches, Difficulty Breathing) | Priority: HIGH

5. Re-Engagement Motivation

User Story	As a survey participant who has not submitted a report in 14 days, I want to receive a personalized reminder explaining why my continued participation matters, so that I feel motivated to re-engage.
LLM Prompt	You are a community health engagement coordinator. A participant last reported on {last_report_date}, which was {days_since} days ago. Their previous symptoms included {last_symptoms}. Write a 3–4 sentence re-engagement message that: (1) acknowledges their prior contribution, (2) explains how their data contributes to community health surveillance, (3) notes any relevant community health developments since their last report, and (4) includes a direct call to action. Use a warm, encouraging tone.
Objective	Reduce survey attrition and maintain longitudinal data quality by delivering personalized, context-aware re-engagement communications.
Data Sources / Priority	Sources: EpiHack participation log, Date of Report field, community-level symptom trends | Priority: MEDIUM

6. Symptom Co-occurrence Explanation

User Story	As a survey participant reporting multiple simultaneous symptoms, I want to understand whether my symptom combination is common or unusual, so that I can assess the urgency of seeking care.
LLM Prompt	You are a clinical epidemiology communicator. The participant reports the following simultaneous symptoms: {symptom_list}. Community data shows this exact combination occurs in {co_occurrence_rate}% of reporters in the past 30 days. In 3 sentences, (1) describe whether this pattern is common or rare, (2) explain what syndromic clusters it may resemble (without diagnosing), and (3) advise on the appropriate level of care to seek based on symptom severity. Note if any symptom (e.g., Difficulty Breathing, Bleeding) warrants immediate evaluation.
Objective	Communicate the clinical significance of multi-symptom profiles to guide participants toward appropriate levels of care.
Data Sources / Priority	Sources: EpiHack co-occurrence matrix, symptom_report.py, clinical triage heuristics | Priority: HIGH

7. Vaccination and Prevention Guidance

User Story	As a survey participant reporting respiratory symptoms, I want to know what preventive measures and vaccinations may be relevant, so that I can protect myself and my household.
LLM Prompt	You are a preventive health advisor. The participant reports {respiratory_symptoms} in postal area {postal_code}. Respiratory illness rates in this area are {direction} compared to the 30-day baseline by {pct_change}%. In 3–4 sentences, describe relevant preventive measures (hand hygiene, masking, ventilation), reference any applicable vaccination recommendations (e.g., influenza, COVID-19, RSV), and advise them to consult their healthcare provider for personalized guidance. Do not make specific vaccination mandates.
Objective	Deliver targeted preventive health guidance linked to current community respiratory disease trends.

Data Sources / Priority	Sources: baseline_change_report outputs, ADHS immunization schedules, ASIIS (AZ Immunization Info System) | Priority: MEDIUM

8. Animal Exposure Risk Screening

User Story	As a survey participant who has reported fever and rash, I want to know if I should mention recent animal contact to my doctor, so that zoonotic causes are not overlooked.
LLM Prompt	You are a One Health clinical communicator. The participant reports Fever and Rash. Prompt them with 3–4 plain-language questions to assess potential zoonotic exposure: (1) recent contact with livestock, poultry, or wildlife; (2) recent tick or mosquito bites; (3) occupation involving animal contact; (4) recent travel to rural or agricultural areas. Then explain in 2 sentences why disclosing animal contact to their physician matters for differential diagnosis. Do not alarm the participant.
Objective	Screen for zoonotic exposure pathways by integrating animal contact history into participant-level risk communication within the One Health framework.
Data Sources / Priority	Sources: USDA APHIS, AZ Dept Agriculture, ArboNET, EpiHack Occupation field, Great AZ Tick Check | Priority: HIGH

9. Household Member Risk Assessment

User Story	As a survey participant, I want to know if other members of my household should also submit symptom reports, so that we can collectively contribute to surveillance and protect each other.
LLM Prompt	You are a household health coordinator. The participant (Household Member ID: {hh_id}) has reported {symptoms}. In 3 sentences, explain the importance of household-level surveillance for detecting secondary transmission, provide instructions for enrolling household members in EpiHack, and describe what symptoms should prompt a new household member to submit an urgent report.
Objective	Extend surveillance coverage to household clusters, enabling secondary attack rate estimation and household transmission chain analysis.
Data Sources / Priority	Sources: EpiHack Household Member ID field, community transmission rate estimates | Priority: MEDIUM

10. Wastewater Surveillance Contextualization

User Story	As a survey participant, I want to understand how my individual report connects to community-wide disease monitoring, so that I appreciate the value of my participation.
LLM Prompt	You are a public health data engagement communicator. In 3–4 sentences, explain to a community member how their individual symptom report is one data stream within a multi-source surveillance system. Describe (in lay terms) how wastewater surveillance, emergency department visits, and community reports like theirs are triangulated to detect outbreaks. Reference the current wastewater trend for {postal_area_region} if available: {wastewater_signal}. Conclude with an affirmation of the participant's contribution.
Objective	Build participant trust and sustained engagement by communicating the collective value of participatory surveillance data within a multi-stream One Health system.
Data Sources / Priority	Sources: ADHS Wastewater Surveillance, BioSense/NSSP, EpiHack participation data | Priority: MEDIUM

11. Loss of Smell or Taste Signal

User Story	As a survey participant reporting loss of smell or taste, I want to understand the epidemiological significance of this symptom, so that I can make an informed decision about testing.
LLM Prompt	You are a clinical epidemiology advisor. The participant reports Loss of Smell or Taste. In the past 7 days, {loss_of_smell_rate}% of community reporters in {postal_code} have reported this symptom (baseline: {baseline_rate}%). In 3 sentences, explain the differential diagnoses associated with this symptom cluster in current surveillance context, recommend appropriate testing options (without mandating), and advise on isolation precautions pending test results.

Objective	Deliver symptom-specific epidemiological context for loss of smell/taste to guide testing and isolation decisions.
Data Sources / Priority	Sources: baseline_change_report.py (smell symptom), EpiHack community rates, ADHS testing site locator | Priority: HIGH

12. Demographic Risk Stratification Feedback

User Story	As an older adult (age 61+) survey participant, I want to know if my age group is disproportionately affected by current illness trends, so that I can take additional precautions if needed.
LLM Prompt	You are an age-stratified risk communicator. The participant is {age} years old (group: 61+). In the past {N} days, {age_group_rate}% of 61+ reporters in {postal_code} reported symptoms, compared to {overall_rate}% across all age groups. In 3–4 sentences, explain whether older adults in the community are disproportionately affected, describe the specific symptoms driving this pattern, and provide tailored guidance for high-risk older adults. Use respectful, empowering language.
Objective	Deliver age-stratified risk communication to high-vulnerability demographic groups based on current community symptom prevalence patterns.
Data Sources / Priority	Sources: symptom_report.py age bins (61+), EpiHack Age field, AZSVI vulnerability scores | Priority: HIGH

13. Mental Health and Illness Burden

User Story	As a survey participant who has been reporting symptoms for several consecutive weeks, I want to receive acknowledgment of the mental health impact of prolonged illness, so that I feel supported and know where to seek help.
LLM Prompt	You are a compassionate health communicator. The participant has reported symptoms continuously for {consecutive_weeks} weeks. In 3–4 sentences: (1) acknowledge the physical and emotional burden of prolonged illness; (2) normalize the psychological impact (anxiety, fatigue, isolation) without pathologizing; (3) provide 1–2 concrete resources for mental health support available in Arizona (e.g., 211 Arizona, behavioral health warmlines); (4) encourage them to discuss persistent symptoms with their provider. Use warm, non-clinical language.
Objective	Address the mental health co-burden of prolonged illness reporting and connect participants to behavioral health resources within the surveillance platform.
Data Sources / Priority	Sources: 211 Arizona, AzHAN behavioral health resources, EpiHack longitudinal report history | Priority: MEDIUM

14. Occupational Exposure Risk

User Story	As a healthcare worker or essential worker survey participant, I want to understand if my occupation places me at elevated risk for the symptoms currently circulating, so that I can take appropriate workplace precautions.
LLM Prompt	You are an occupational health epidemiologist. The participant's occupation is {occupation}. Current community surveillance shows {high_symptom} is elevated by {pct_change}% above baseline in {postal_code}. In 3–4 sentences, explain how the participant's occupational context may affect their exposure risk, describe appropriate occupational health precautions (PPE, reporting to occupational health), and recommend whether they should report symptoms to their employer's occupational health program.
Objective	Integrate occupational exposure context into individual risk communication for high-risk worker populations within the EpiHack surveillance cohort.
Data Sources / Priority	Sources: EpiHack Occupation field, ADHS occupational health guidance, community symptom rates | Priority: HIGH

15. Rash and Bleeding Symptom Urgency Triage

User Story	As a survey participant reporting rash or bleeding from body openings, I want to know whether these symptoms require emergency evaluation, so that I can make a timely decision about seeking care.
LLM Prompt	You are an emergency health triage advisor. The participant reports {urgent_symptoms} (Rash and/or Bleeding from Body Openings). These symptoms may indicate conditions requiring urgent evaluation. In 2–3 sentences: (1) clearly state that these symptoms warrant prompt clinical evaluation; (2) specify emergency warning signs that require calling 911 or going to an emergency department immediately; (3) provide the nearest Arizona emergency health contact. Use a calm but direct tone. Do not attempt to diagnose.
Objective	Implement automated clinical triage escalation for high-acuity symptom combinations reported through the EpiHack participatory surveillance system.
Data Sources / Priority	Sources: EpiHack symptom flags (Rash, Bleeding), ADHS emergency guidance, 24/7 Nurse Hotline | Priority: CRITICAL

16. Progress Feedback and Gamification

User Story	As a long-term survey participant, I want to see a summary of my contribution to the EpiHack community health network over time, so that I feel recognized for my sustained engagement.
LLM Prompt	You are a community health engagement system. The participant has submitted {total_reports} reports over {months_active} months. In 3–4 sentences: (1) summarize their total contribution quantitatively; (2) describe how their data contributed to at least one community-level insight (e.g., 'Your reports during July helped detect an emerging fever cluster in your area'); (3) award them a participation milestone and invite them to encourage a neighbor to join. Use enthusiastic, community-affirming language.
Objective	Sustain longitudinal participant engagement through contribution recognition and social network activation, improving data quality and coverage.
Data Sources / Priority	Sources: EpiHack participation log, community signal detection history | Priority: MEDIUM

17. Pediatric Household Symptom Reporting

User Story	As a parent survey participant, I want guidance on reporting symptoms for children under 18 in my household, so that pediatric illness patterns are captured in the surveillance system.
LLM Prompt	You are a family health surveillance coordinator. The participant is reporting on behalf of a child (age: {child_age}). In 3–4 sentences: (1) explain the specific symptom flags most clinically significant in pediatric populations (Fever, Rash, Difficulty Breathing, Diarrhea); (2) describe when pediatric symptoms require emergency evaluation vs. telehealth consultation; (3) explain how pediatric reports are handled in the surveillance system with appropriate privacy protections.
Objective	Extend EpiHack surveillance to pediatric household members through parent-proxy reporting, capturing age-stratified disease burden in the community.
Data Sources / Priority	Sources: EpiHack Household Member ID, pediatric clinical triage thresholds, ADHS child health guidelines | Priority: HIGH

18. Spanish-Language Health Communication

User Story	As a Spanish-speaking survey participant, I want to receive health information in my primary language, so that I can fully understand my risk and take appropriate action.
LLM Prompt	You are a bilingual public health communicator. Translate and culturally adapt the following health message into plain Spanish (8th-grade reading level or below), ensuring medical terms are explained in everyday language: {english_health_message}. Use culturally appropriate examples relevant to Arizona's Latino community. Avoid direct word-for-word translation; prioritize comprehension and cultural relevance. End with a sentence encouraging the participant to share the information with their family.
Objective	Ensure health-literacy-appropriate, culturally adapted risk communication reaches Spanish-speaking participants in Arizona's Latino communities.

Data Sources / Priority	Sources: EpiHack participant language preference, ADHS Spanish-language resources, 211 Arizona bilingual services | Priority: HIGH
19. Tribal Community Health Messaging	
User Story	As a survey participant from a tribal community, I want to receive health information that respects my community's data sovereignty and cultural context, so that the information feels relevant and trustworthy.
LLM Prompt	You are a culturally responsive health communicator working with Indigenous communities. The participant is from {tribal_community} in Arizona. Draft a 3–4 sentence health message that: (1) respects tribal health governance and data sovereignty; (2) connects current surveillance findings to the community's specific environmental and cultural context; (3) references the appropriate tribal health department as the primary point of contact; (4) avoids paternalistic framing. Do not share individual or aggregate data without confirming appropriate tribal data governance authorization.
Objective	Deliver culturally responsive, sovereignty-respecting health communication to tribal community participants within the One Health participatory surveillance framework.
Data Sources / Priority	Sources: Tribal health departments (22 AZ tribes), USDA APHIS tribal liaisons, tribal mosquito/vector surveillance | Priority: CRITICAL
20. Post-Recovery Follow-Up	
User Story	As a survey participant who has recovered from reported symptoms, I want to receive a follow-up message confirming my recovery contributes to surveillance and asking about any lingering effects, so that I feel the platform values my complete health journey.
LLM Prompt	You are a longitudinal health surveillance coordinator. The participant last reported symptoms {symptom_list} on {illness_date} and has since reported No Symptoms for {recovery_days} consecutive days. In 3–4 sentences: (1) acknowledge their recovery; (2) ask 1–2 brief follow-up questions about lingering effects (e.g., fatigue, brain fog) relevant to post-acute sequelae; (3) explain how recovery data improves the platform's understanding of illness duration and burden; (4) invite them to continue reporting for at least 30 more days.
Objective	Capture post-acute illness data through structured follow-up prompts, enabling illness duration and post-acute sequelae surveillance within EpiHack.
Data Sources / Priority	Sources: EpiHack longitudinal report history, No Symptoms flag, Date of Report field | Priority: MEDIUM

Section 3: LLM-Ready User Story Prompts — Public Health Officials

The following 20 user story prompts are designed for public health officials, epidemiologists, and surveillance analysts using EpiHack data. Each prompt enables LLM-mediated synthesis of multi-source One Health data into actionable operational intelligence products.

1. Outbreak Signal Summary Brief	
User Story	As a state epidemiologist, I want a concise daily situation report summarizing emerging symptom signals across all Arizona postal areas, so that I can prioritize outbreak investigations.
LLM Prompt	You are an epidemiological intelligence analyst. Based on the following EpiHack signal table for the past 7 days: {signal_table} (columns: postal_code, symptom, recent_rate, baseline_rate, pct_change, signal_class). Produce a structured situation report with: (1) top 5 signals by magnitude (signal='up' or 'new'); (2) postal areas with 3+ elevated signals; (3) any newly emerging symptoms (signal='new'); (4) a recommended investigation priority list. Format as a bulleted executive brief. Flag small sample sizes (n < 10) explicitly.
Objective	Automate daily synthesis of EpiHack surveillance signals into a prioritized operational situation report for state epidemiologists.
Data Sources / Priority	Sources: baseline_change_report.py, EpiHack signal classification (up/dn/new/nc/indef), ADHS MEDSIS | Priority: CRITICAL
2. Geographic Cluster Detection	
User Story	As a surveillance analyst, I want to identify postal code clusters where multiple symptoms are simultaneously elevated, so that I can detect potential outbreak foci for field investigation.
LLM Prompt	You are a spatial epidemiology analyst. Given the following postal code × symptom signal matrix: {postal_symptom_matrix}. Identify postal codes where 3 or more symptoms show 'up' or 'new' signals in the past 7 days. For each identified cluster: (1) list the elevated symptoms and their % change from baseline; (2) calculate the composite burden score (sum of pct_changes); (3) rank clusters by composite burden. Output as a structured table. Note any postal codes with insufficient data (signal='indef').
Objective	Identify multi-symptom geographic clusters in participatory surveillance data to prioritize field investigation resources.
Data Sources / Priority	Sources: baseline_change_report.py postal×symptom heatmap, EpiHack Geographical Coordinates, symptom_report.py | Priority: CRITICAL
3. One Health Zoonotic Risk Assessment	
User Story	As a One Health surveillance coordinator, I want a risk narrative integrating human symptom signals with animal disease reports and environmental conditions, so that I can assess zoonotic spillover probability.
LLM Prompt	You are a One Health risk intelligence analyst. Integrate the following data streams: Human surveillance: {human_signals}. Animal reports: {animal_signals} (source: USDA APHIS, AZ Dept Agriculture). Environmental conditions: {environmental_data} (NWS HeatRisk, ADHS Wastewater). Vector activity: {vector_data} (ArboNET, Great AZ Tick Check). In 400–500 words, produce a One Health risk narrative that: (1) identifies symptom patterns consistent with zoonotic etiology; (2) correlates human signals with animal/environmental/vector data; (3) assigns a zoonotic spillover risk level (Low/Medium/High/Critical); (4) recommends cross-sector response actions.
Objective	Generate integrated One Health risk narratives by synthesizing human, animal, environmental, and vector surveillance streams for zoonotic threat assessment.
Data Sources / Priority	Sources: USDA APHIS, AZ Dept Agriculture, ArboNET, NWS HeatRisk, ADHS Wastewater, EpiHack human signals | Priority: CRITICAL
4. Health Equity Surveillance Report	

User Story	As a health equity officer, I want to identify whether symptom burden disproportionately affects socially vulnerable communities, so that I can target interventions to populations with the greatest need.
LLM Prompt	You are a health equity epidemiologist. Given symptom prevalence rates by postal code: {postal_rates} and AZSVI vulnerability scores by postal code: {azsvi_scores}. Conduct a correlation analysis between symptom burden and social vulnerability. In 3–4 paragraphs: (1) identify the top 5 postal codes with high symptom burden AND high vulnerability (AZSVI quintile 4–5); (2) describe the specific symptoms driving disparity; (3) recommend targeted intervention strategies (mobile clinics, community health workers, translated communications); (4) flag any data gaps that limit equity analysis.
Objective	Identify and communicate health equity dimensions of community symptom burden by integrating EpiHack surveillance data with AZSVI vulnerability stratification.
Data Sources / Priority	Sources: AZSVI, SVI (CDC), EpiHack postal-level rates, symptom_report.py, MySidewalk | Priority: HIGH

5. Wastewater-Clinical-Community Triangulation

User Story	As a surveillance epidemiologist, I want a triangulated signal assessment combining wastewater, emergency department, and community-reported data, so that I can assess signal reliability before launching a public health response.
LLM Prompt	You are a multi-source surveillance triangulation analyst. Synthesize: (1) EpiHack community signals: {epihack_signals}; (2) ADHS Wastewater Surveillance trend for {region}: {wastewater_trend}; (3) BioSense/NSSP emergency department syndromic data: {ed_data}. For each symptom/syndrome with concordant signals across 2+ sources: (1) describe the concordance level; (2) assign a signal confidence tier (Tier 1: all 3 concordant; Tier 2: 2 concordant; Tier 3: single source); (3) recommend appropriate response level per tier. Output as a structured decision table.
Objective	Implement multi-source signal triangulation to improve surveillance specificity and reduce false-positive outbreak responses.
Data Sources / Priority	Sources: ADHS Wastewater Surveillance, BioSense/NSSP, ESSENCE, EpiHack baseline_change_report | Priority: CRITICAL

6. Public Health Communication Draft

User Story	As a public health communications officer, I want an LLM-drafted community health advisory based on current surveillance signals, so that I can rapidly produce accurate, equity-aware public messaging.
LLM Prompt	You are a public health communications specialist. Current EpiHack signals show: {top_signals}. Draft a community health advisory (300–400 words) that: (1) describes current illness trends in plain language (8th-grade reading level); (2) provides actionable recommendations for community members; (3) identifies high-risk groups requiring special attention; (4) includes information on local testing, vaccination, and care resources; (5) uses an empowering, non-stigmatizing tone. Provide both English and Spanish versions.
Objective	Automate equity-aware community health advisory drafting from real-time EpiHack surveillance signals, reducing communications officer workload during outbreak response.
Data Sources / Priority	Sources: EpiHack signals, ADHS communications templates, 211 Arizona, ADHS testing/vaccination locator | Priority: HIGH

7. Demographic Shift Detection Alert

User Story	As an epidemiological analyst, I want to receive an alert when symptom burden shifts significantly toward a specific age or sex group, so that I can investigate potential demographic-specific outbreaks or reporting biases.
LLM Prompt	You are a demographic surveillance analyst. Compare symptom prevalence by age group and sex for the current 7-day window vs. the prior 30-day baseline: {demographic_comparison_table}. Identify any age×sex strata where: (1) symptom rates have increased by >20% relative to baseline; (2) the stratum's share of total reporters has shifted by >10 percentage points. For each identified shift:

	describe the pattern, assess whether it likely reflects a true epidemiological signal or a reporting artifact, and recommend appropriate follow-up actions.
Objective	Detect demographically concentrated symptom signals that may indicate age- or sex-specific disease dynamics or differential healthcare access patterns.
Data Sources / Priority	Sources: symptom_report.py age×sex stratification, EpiHack Age and Sex fields, ADHS demographic data | Priority: HIGH
8. Rare Symptom Cluster Investigation	
User Story	As a state epidemiologist, I want to be alerted when unusual symptom combinations appear in the community data, so that I can investigate potential sentinel events for rare or novel pathogens.
LLM Prompt	You are an epidemiological signal analyst specializing in rare event detection. Review the following co-occurrence matrix of symptoms reported in the past {N} days: {co_occurrence_matrix}. Flag any symptom combinations that: (1) have a prevalence rate >2 standard deviations above the 90-day historical mean; (2) include high-concern symptoms (Bleeding from Body Openings, Yellow Skin/Eyes, Discolored Urine); (3) cluster in fewer than 3 postal codes. For each flagged combination, generate a differential diagnosis list (top 5 candidate syndromes) and recommend confirmatory laboratory or epidemiological investigations.
Objective	Generate automated sentinel event alerts for rare or high-concern symptom combinations that may indicate novel pathogen emergence or unusual disease clusters.
Data Sources / Priority	Sources: EpiHack 14-symptom flags, baseline_change_report signal matrix, ProMED, ADHS MEDSIS | Priority: CRITICAL
9. Reporter Coverage and Bias Assessment	
User Story	As a surveillance data quality officer, I want to assess whether EpiHack participant coverage is representative of the underlying community population, so that I can quantify and communicate surveillance biases.
LLM Prompt	You are a surveillance epidemiology methodologist. Given: EpiHack reporter demographics: {reporter_demographics} (age, sex, postal code distribution). Reference population: {census_demographics} (ACS 5-year estimates for covered postal codes). In 3–4 paragraphs: (1) quantify over- and under-represented demographic groups; (2) estimate the potential direction of bias in reported symptom rates for under-represented groups; (3) recommend weighting strategies or targeted recruitment to improve representativeness; (4) calculate a coverage score (EpiHack reporters / estimated population) by postal code.
Objective	Assess and communicate demographic representativeness of the EpiHack participatory surveillance cohort to quantify potential surveillance bias.
Data Sources / Priority	Sources: EpiHack Age, Sex, Postal Code fields, US Census ACS, MySidewalk demographic data | Priority: HIGH
10. Seasonal Baseline Recalibration	
User Story	As a surveillance analyst, I want to recalibrate symptom baselines to account for seasonal variation, so that summer heat season signals are not confounded by expected seasonal patterns.
LLM Prompt	You are an epidemiological statistician. Given 12 months of EpiHack symptom data: {annual_symptom_timeseries}. For each of the 14 symptom flags: (1) fit a seasonal decomposition model (additive or multiplicative, specify); (2) extract the seasonal component; (3) calculate a seasonally adjusted baseline for the current month; (4) identify symptoms where the current 7-day rate exceeds the seasonally adjusted baseline by >15%. Present results as a structured table with columns: symptom, raw_rate, seasonal_expected, adjusted_pct_change, signal. Flag months with insufficient reporter counts (n < 20).
Objective	Implement seasonal baseline adjustment for EpiHack symptom surveillance to improve signal specificity by removing expected seasonal fluctuations.

Data Sources / Priority	Sources: EpiHack 12-month symptom timeseries, NOAA climate data, statistical decomposition (STL/X-13) | Priority: HIGH
11. Vector-Borne Disease Correlation	
User Story	As an arboviral disease epidemiologist, I want to correlate community-reported fever and rash patterns with mosquito trap data and ArboNET reports, so that I can assess arboviral transmission risk.
LLM Prompt	You are an arboviral surveillance analyst. Human data: EpiHack reports of Fever+Rash in {postal_codes} over past {N} days — rate: {fever_rash_rate}%, pct_change: {pct_change}%. Vector data: mosquito trap positivity rates for {area}: {trap_data} (source: Great AZ Mosquito Hunt). ArboNET reported cases: {arbonet_cases}. In 3–4 paragraphs: (1) assess concordance between human symptom signals and vector activity; (2) estimate arboviral risk level for the affected postal codes; (3) recommend enhanced surveillance actions (increased trapping, targeted testing, physician alerts); (4) identify data gaps (e.g., no tick surveillance data for AZ).
Objective	Integrate community-reported symptom signals with vector surveillance data to generate actionable arboviral disease risk assessments under the One Health framework.
Data Sources / Priority	Sources: ArboNET, Great AZ Mosquito Hunt, Great AZ Tick Check, EpiHack Fever+Rash signals, ADHS Mosquito-Borne Disease | Priority: CRITICAL
12. Emergency Resource Allocation Recommendation	
User Story	As a public health emergency preparedness officer, I want data-driven recommendations for deploying mobile testing units and surge resources, so that I can direct resources to the highest-need postal areas.
LLM Prompt	You are a public health operations analyst. Given EpiHack signal data: {high_burden_postal_codes} (symptom rates, % change, AZSVI vulnerability scores, reporter counts). Available resources: {available_resources} (mobile testing units, community health workers, PPE caches). In 3–4 paragraphs: (1) rank postal codes by composite need score (symptom burden × vulnerability × population density); (2) allocate available resources to the top-ranked areas with justification; (3) identify postal codes with insufficient EpiHack data where resource allocation should rely on alternative data sources; (4) propose a monitoring cadence for resource deployment effectiveness.
Objective	Generate evidence-based emergency resource allocation recommendations by integrating EpiHack symptom signals with social vulnerability and resource availability data.
Data Sources / Priority	Sources: EpiHack postal signals, AZSVI, Census population data, ADHS emergency preparedness inventory | Priority: CRITICAL
13. School and Childcare Illness Monitoring	
User Story	As a school health surveillance coordinator, I want to track symptom trends among school-age participants, so that I can issue timely school health advisories and coordinate with school nurses.
LLM Prompt	You are a school health epidemiologist. EpiHack data for participants age 5–17 in the past {N} days: {pediatric_data} (symptom rates by age group and postal code). In 3–4 paragraphs: (1) identify symptoms disproportionately elevated in school-age groups relative to the adult population; (2) assess whether patterns are consistent with school transmission clusters; (3) draft a 150-word school health advisory for distribution to school principals and nurses; (4) recommend reporting thresholds for triggering school closure notifications.
Objective	Monitor school-age symptom trends in EpiHack data to enable timely school health advisories and coordinate outbreak response with educational institutions.
Data Sources / Priority	Sources: EpiHack Age field (5–17 proxy via pediatric household reporting), ADHS school health guidelines | Priority: HIGH
14. Agricultural Worker Surveillance	

User Story	As a migrant health surveillance officer, I want to monitor symptom trends among agricultural worker participants in rural Arizona postal areas, so that I can detect occupational illness clusters in this high-exposure, underserved population.
LLM Prompt	You are a migrant and agricultural health epidemiologist. EpiHack data filtered to Occupation='Agricultural Worker' and rural postal codes {rural_postal_codes}: {agricultural_worker_data}. Environmental data: {heat_risk_data} (NWS HeatRisk), pesticide exposure reports (if available). In 3–4 paragraphs: (1) describe current symptom burden in this occupational subgroup; (2) identify symptoms consistent with heat illness or pesticide exposure; (3) compare rates to the general community; (4) recommend bilingual (English/Spanish) targeted outreach and screening protocols for agricultural worker health programs.
Objective	Detect occupational illness clusters among agricultural workers by integrating EpiHack occupational data with environmental exposure and health equity stratifiers.
Data Sources / Priority	Sources: EpiHack Occupation field, NWS HeatRisk, ADHS Environmental Health, Migrant Health programs, AZSVI | Priority: HIGH

15. Tribal Nation Health Situation Report

User Story	As a tribal health department epidemiologist, I want a surveillance situation report restricted to tribal community participants that respects data sovereignty, so that I can brief tribal leadership with community-specific health intelligence.
LLM Prompt	You are a tribal public health data analyst. IMPORTANT: This analysis must be restricted to participants who have provided consent for tribal-level data sharing under the relevant tribal data governance agreement. EpiHack data for {tribal_community} participants: {tribal_data}. In 3–4 paragraphs produce a situation report that: (1) summarizes current symptom burden; (2) identifies any emerging signals; (3) contextualizes findings with locally relevant environmental and seasonal factors; (4) recommends community-specific response actions. All outputs must be reviewed by the tribal health director before external distribution. Do not publish individual-level data.
Objective	Produce sovereignty-respecting, community-specific health situation reports for tribal nation public health officials using EpiHack participatory data.
Data Sources / Priority	Sources: Tribal health departments (22 AZ tribes), tribal data governance agreements, EpiHack tribal participant subset | Priority: CRITICAL

16. LLM-Assisted Epidemiological Hypothesis Generation

User Story	As a research epidemiologist, I want the LLM to generate testable epidemiological hypotheses from current surveillance patterns, so that I can design targeted investigations and studies.
LLM Prompt	You are an epidemiological hypothesis generator. Current EpiHack surveillance findings: {surveillance_summary}. One Health context: {one_health_context}. Generate 5 testable epidemiological hypotheses that could explain the observed patterns. For each hypothesis: (1) state the hypothesis in formal epidemiological language (exposure, outcome, population, time); (2) identify the study design best suited to test it (cohort, case-control, cross-sectional, ecological); (3) list required data sources; (4) estimate feasibility (data availability score: 1–5). Prioritize hypotheses with convergent support from multiple data streams.
Objective	Leverage LLM reasoning to generate structured, testable epidemiological hypotheses from multi-source One Health surveillance data, accelerating outbreak investigation design.
Data Sources / Priority	Sources: EpiHack all data streams, MEDSIS, BioSense/NSSP, ArboNET, USDA APHIS, ADHS Wastewater | Priority: HIGH

17. After-Action Report Synthesis

User Story	As a public health emergency manager, I want an after-action report summarizing EpiHack's surveillance performance during a recent outbreak event, so that I can identify system strengths and gaps for continuous quality improvement.
LLM Prompt	You are a public health systems evaluator. Summarize EpiHack's surveillance performance during the period {outbreak_period} for {outbreak_event}. Use the following performance metrics: {metrics} (reporter counts, signal lead time vs. MEDSIS confirmation, geographic coverage, demographic

	representativeness, false positive rate). In 4–5 paragraphs: (1) describe the system's early warning performance; (2) identify geographic and demographic coverage gaps; (3) assess signal specificity vs. laboratory-confirmed cases; (4) recommend 3–5 specific system improvements; (5) quantify the public health value of early community signal detection.
Objective	Generate structured after-action reports evaluating EpiHack participatory surveillance performance relative to traditional surveillance systems for continuous quality improvement.
Data Sources / Priority	Sources: EpiHack historical data, MEDSIS confirmed cases, BioSense/NSSP, ESSENCE, ADHS outbreak records | Priority: HIGH
18. Media and Infodemic Monitoring	
User Story	As a public health communications officer, I want to monitor whether community symptom reports correlate with media-driven health anxiety cycles, so that I can distinguish true epidemiological signals from infodemic artifacts.
LLM Prompt	You are a public health infodemic analyst. EpiHack reporter counts and symptom flags over the past 30 days: {epihack_timeseries}. Media event log: {media_events} (dates of major health news stories, social media spikes). In 3–4 paragraphs: (1) identify temporal correlations between media events and EpiHack report submission surges; (2) distinguish symptom clusters that preceded media coverage (true signals) from those that followed it (potential infodemic artifacts); (3) recommend reporting language adjustments to reduce infodemic amplification; (4) propose a media correlation monitoring protocol for routine surveillance.
Objective	Distinguish true epidemiological signals from media-driven health anxiety artifacts in participatory surveillance data to improve signal specificity.
Data Sources / Priority	Sources: EpiHack participation timeseries, ProMED, social media trend data (Meta/Google Trends), ADHS communications log | Priority: MEDIUM
19. Predictive Risk Modeling Brief	
User Story	As a state health data scientist, I want an LLM-synthesized interpretation of a machine learning–based disease risk forecast, so that I can communicate model outputs to non-technical public health leadership.
LLM Prompt	You are a public health data science communicator. A gradient boosting model predicting 14-day disease risk scores by postal code has produced the following outputs: {model_predictions} (postal_code, risk_score, top_3_features, confidence_interval). In 3–4 paragraphs: (1) explain the model's top 3 most important predictive features in plain language; (2) identify the 5 postal codes with the highest predicted risk; (3) communicate model uncertainty using confidence intervals in language appropriate for non-technical health leadership; (4) recommend which high-risk areas should receive field verification given model limitations.
Objective	Bridge the gap between ML-generated disease risk predictions and public health decision-making by generating plain-language model interpretation reports for leadership audiences.
Data Sources / Priority	Sources: EpiHack ML risk model outputs, AZSVI, Census demographic covariates, historical EpiHack timeseries | Priority: HIGH
20. Legislative and Policy Intelligence Brief	
User Story	As a state health policy advisor, I want a policy brief summarizing the implications of current surveillance trends for public health legislation and resource appropriation, so that I can inform legislative decision-making.
LLM Prompt	You are a public health policy analyst. Current EpiHack and One Health surveillance context: {surveillance_summary}. Arizona legislative priorities: {legislative_context}. In 500–600 words, produce a policy intelligence brief that: (1) summarizes current disease burden trends in 2–3 accessible sentences; (2) identifies 3 policy levers that could reduce identified health risks (e.g., heat illness prevention legislation, expanded tribal surveillance funding, wastewater monitoring mandate); (3) estimates the public health ROI of each policy lever with available evidence; (4) recommends 2–3 specific legislative asks with supporting surveillance data. Use formal policy brief format with an executive summary.

Objective	Generate evidence-based policy intelligence briefs synthesizing EpiHack and One Health surveillance data for state health policy advisors and legislative audiences.
Data Sources / Priority	Sources: EpiHack all signals, AZSVI, ADHS annual report, AZ Legislative priorities, CDC policy frameworks | Priority: HIGH

Section 4: Additional Considerations for LLM Integration

4.1 Privacy, De-identification, and HIPAA Compliance

All LLM prompt templates that incorporate individual-level participant data must enforce minimum cell size thresholds (recommended: $n \geq 5$ for geographic/demographic aggregates) before generating community-level rates. PII fields (Email, Phone Number, Unique ID) must be stripped from any data passed to LLM APIs. If using third-party LLM providers (e.g., OpenAI, Anthropic API), a HIPAA Business Associate Agreement (BAA) must be in place. Prompt templates should explicitly instruct the LLM to avoid re-identifying individuals from contextual clues.

4.2 Health Literacy and Plain Language Standards

LLM outputs directed at community participants should target a 6th–8th grade reading level (Flesch-Kincaid or SMOG index). Prompt templates should instruct the LLM to avoid medical jargon, explain technical terms parenthetically when unavoidable, and use active voice. All participant-facing templates should be tested for readability using automated scoring tools before deployment. Spanish-language templates should be developed by bilingual public health professionals and back-translated for validation.

4.3 Statistical Power and Uncertainty Communication

The current EpiHack dataset (N=100 total records, one record per postal code) yields extremely sparse windows: a 7-day recent window contains only 4 records. LLM prompt templates must be designed to propagate this uncertainty to end users. Specifically: (1) templates should include explicit sample size reporting (e.g., 'based on {n} reporters'); (2) signals from strata with $n < 5$ should be flagged as 'insufficient data' and the LLM should be instructed not to generate rate-based interpretations; (3) all quantitative claims should include confidence language ('approximately', 'preliminary estimates suggest').

4.4 One Health Data Integration Architecture

To fulfill the platform's One Health mandate, EpiHack requires a data integration layer connecting: human syndromic data (EpiHack), animal disease reports (USDA APHIS, AZ Dept Agriculture), environmental sensors (NWS HeatRisk, ADHS Wastewater), and vector surveillance (ArboNET, mosquito/tick trap data). LLM prompts for One Health narratives should be designed to gracefully handle missing data streams by instructing the model to note data gaps explicitly rather than omitting the One Health framing.

4.5 Tribal Data Sovereignty and OCAP® Principles

Data governance for tribal community participants must comply with OCAP® principles (Ownership, Control, Access, Possession) and applicable tribal data sovereignty frameworks. LLM prompt templates involving tribal community data should: (1) require explicit tribal health department authorization before processing; (2) be reviewed by tribal cultural liaisons; (3) not aggregate tribal data with general Arizona data without explicit consent; (4) acknowledge that federally recognized tribes are sovereign nations with independent public health jurisdictions.

4.6 Clinical Safety and LLM Liability Boundaries

EpiHack LLM outputs are intended as public health information tools, not clinical diagnostic or treatment systems. All participant-facing prompts must include a standard disclaimer: 'This information is for community health awareness only and does not constitute medical advice. If you are experiencing a medical emergency, call 911.' High-acuity symptom combinations (Difficulty Breathing, Bleeding, Chest Pain) should trigger mandatory escalation to human clinical review or emergency referral rather than relying solely on LLM-generated triage guidance.

4.7 Temporal Latency and Data Freshness

EpiHack surveillance signals are anchored to the maximum date present in the dataset (currently July 12, 2025 in the fakeData.csv). In production, the surveillance anchor date should be updated in real-time as new participant reports are submitted. LLM prompts should include the data freshness date (e.g., 'based on reports through {max_date}') and instruct the model to acknowledge that rapidly evolving situations may not be fully captured in the current data window.

4.8 LLM Hallucination Mitigation

LLM models may generate plausible-sounding but factually incorrect epidemiological statistics or clinical guidance. To mitigate hallucination risk: (1) all quantitative values in prompts should be passed as structured data parameters rather than relying on the LLM's parametric knowledge; (2) prompts should explicitly instruct the model to base its response only on the provided data and not to extrapolate from training knowledge; (3) all LLM outputs for public health officials should undergo human expert review before distribution; (4) retrieval-augmented generation (RAG) with a curated One Health knowledge base is recommended for production deployment.

Section 5: Clarifying Questions for Platform Development

5.1 Platform Architecture

- Q1. Which LLM provider(s) are planned for the EpiHack backend (e.g., Anthropic Claude, OpenAI GPT-4o, Google Gemini, open-source models via Ollama)? This determines prompt engineering constraints, context window limits, and BAA/data governance requirements.
- Q2. Will LLM prompts be executed server-side (centralized API calls) or client-side (local model inference)? This significantly affects data privacy architecture and latency constraints for real-time surveillance.
- Q3. Is there a planned real-time data ingestion pipeline, or will EpiHack remain a batch-processed surveillance system? Real-time operation requires different temporal anchoring logic than the current dataset-max-date approach.
- Q4. What is the intended data volume at production scale? The current fakeData.csv has 100 records (one per postal code); production Arizona postal coverage would require ≥300 postal codes × potentially thousands of participants, requiring scalable statistical backends.

5.2 Data Availability and Governance

- Q5. Is there a formal data sharing agreement with ADHS for MEDSIS, BioSense/NSSP, and ESSENCE data? These are the critical triangulation sources for validating EpiHack community signals.
- Q6. What is the current data access status for AZSVI postal code linkage? This is essential for equity-stratified analyses and high-priority for the health equity user stories.
- Q7. Are tribal health departments already engaged as partners? Tribal data governance must be established before any tribal-specific surveillance or prompt templates can be deployed.
- Q8. Is the ADHS Wastewater Surveillance data publicly accessible via API, or does it require formal data sharing agreements? The triangulation stories (Official Story #5) depend on this.
- Q9. What is the minimum expected reporter density per postal code per week in production? This determines the feasibility of postal-code-level rate calculations and the appropriate minimum cell size thresholds.

5.3 LLM Configuration and Prompt Engineering

- Q10. Will the LLM operate with a fixed system-role prompt defining its public health scope, or will role instructions be included in each user-level prompt? A global system prompt enforcing safety boundaries is strongly recommended.
- Q11. What is the planned chain-of-thought or reasoning approach? For complex One Health risk narratives (Official Story #3, #11), multi-step reasoning (CoT or ReAct) will significantly improve output quality over single-shot prompting.
- Q12. Is retrieval-augmented generation (RAG) planned? A curated One Health knowledge base (ADHS guidelines, CDC reference materials, Arizona-specific disease profiles) would substantially improve accuracy and reduce hallucination risk.
- Q13. How will LLM output quality be evaluated in production? Recommend establishing automated evaluation metrics: readability score (Flesch-Kincaid), factual consistency (against source data), and clinical safety review for high-acuity outputs.

5.4 Prioritization and Phasing

- Q14. Which user story categories should be prioritized for the initial production release? Recommended phasing: Phase 1 (CRITICAL stories: outbreak signal, emerging alert, zoonotic risk, high-acuity triage); Phase 2 (HIGH: equity, demographic shift, wastewater triangulation); Phase 3 (MEDIUM: engagement, gamification, follow-up).
- Q15. Is there a formal IRB protocol or public health surveillance authority designation covering EpiHack data collection and LLM-based output generation? This determines the regulatory framework for data use, participant consent, and clinical guidance liability.